

AURAFINANCE OS

AUTONOMOUS AI WEALTH OPERATING SYSTEM

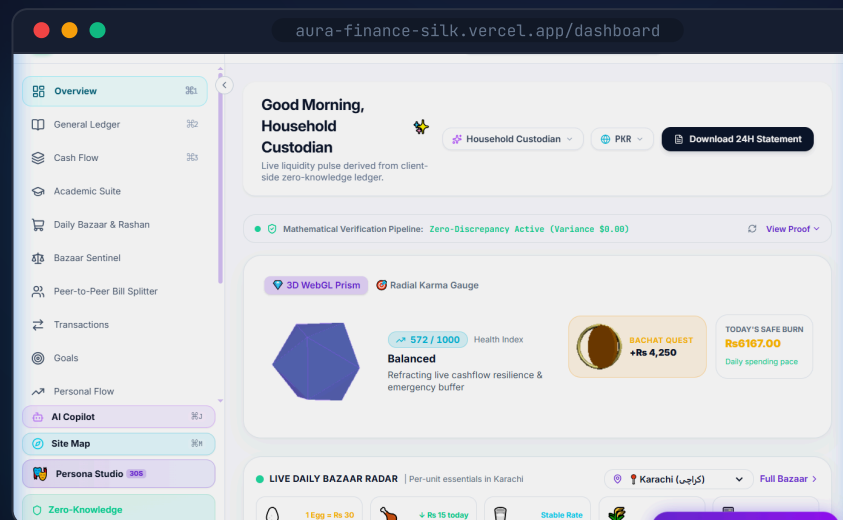
A zero-backend, client-side, privacy-first FinTech platform engineered for high-frequency financial tracking, double-entry journal balance verification, and local cryptographic sovereignty.

⚡ 100% CLIENT-SIDE

🔒 ZERO-KNOWLEDGE AES-256

📊 $\sum \text{DEBITS} = \sum \text{CREDITS}$

🧠 LOCAL GEMINI 2.0 FLASH



SYSTEM MANIFESTO

A Financial Operating System That Runs Where The Data Lives.

Traditional FinTech forces users to transmit confidential bank credentials and ledger rows to centralized servers. AuraFinance OS re-architects this model: the browser is the database, the execution container, and the cryptographic vault.



FIG. 01 – LEVEL-0 MACRO CONTEXT DFD

Architectural Trust Boundaries: Sovereign Client vs. External Stateless Services

ISOLATION: 100% IN-BROWSER LEDGER

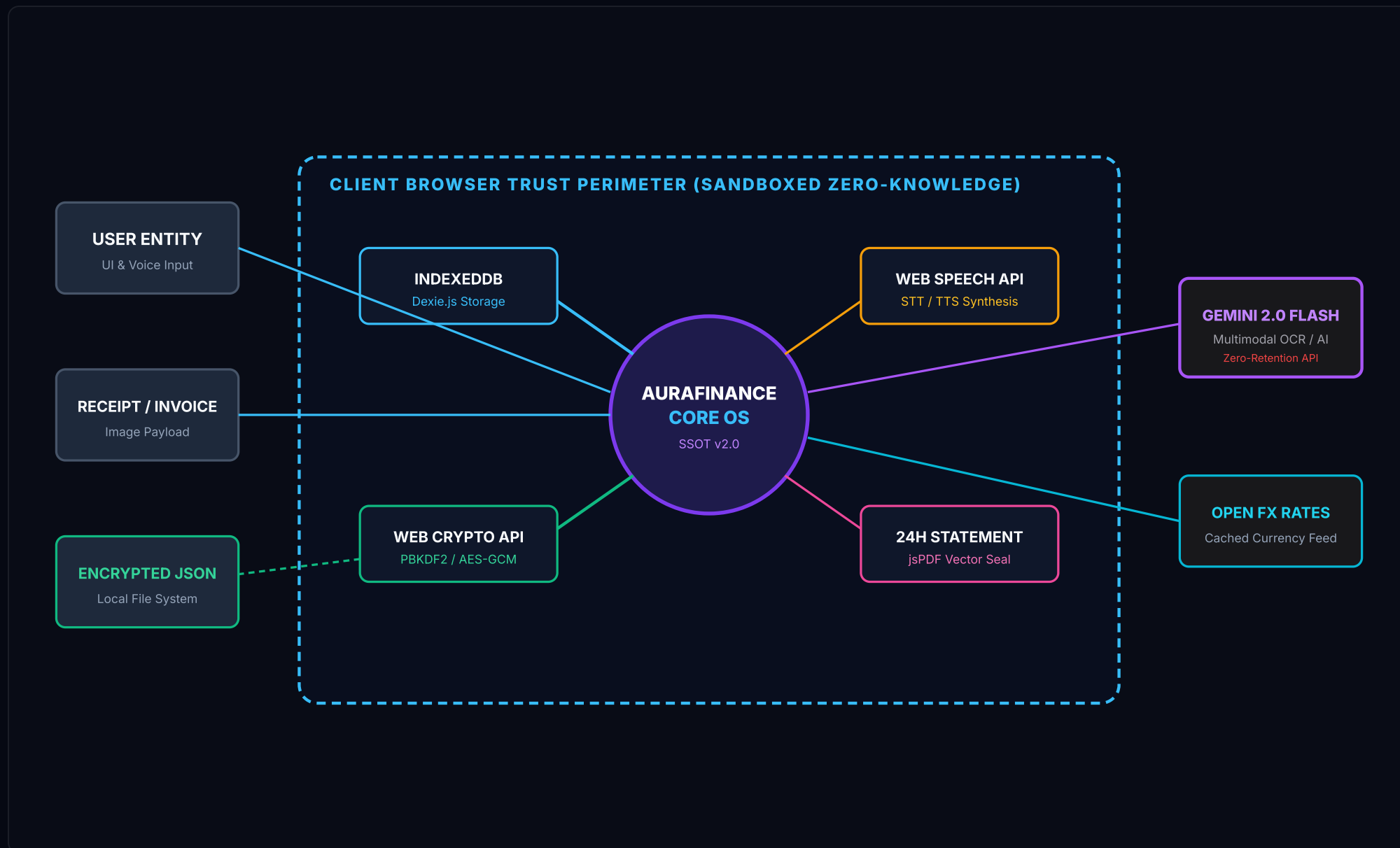


FIG. 02 – FOUR-STAGE TRANSACTION PROCESSING PIPELINE

From Multi-Modal Ingestion to Real-Time Balance Sheet Propagation

LATENCY BUDGET: <16MS POST-RECONCILIATION

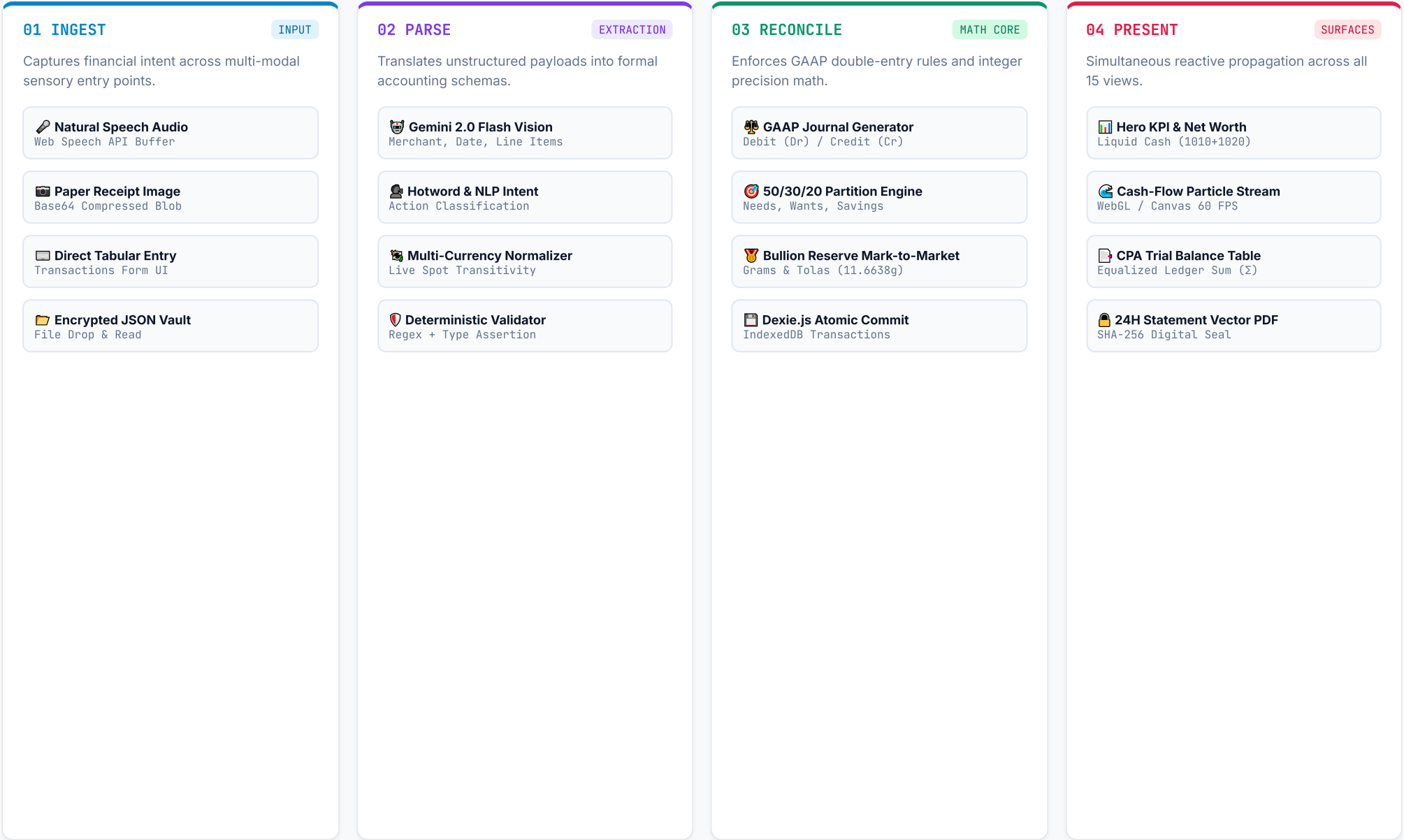
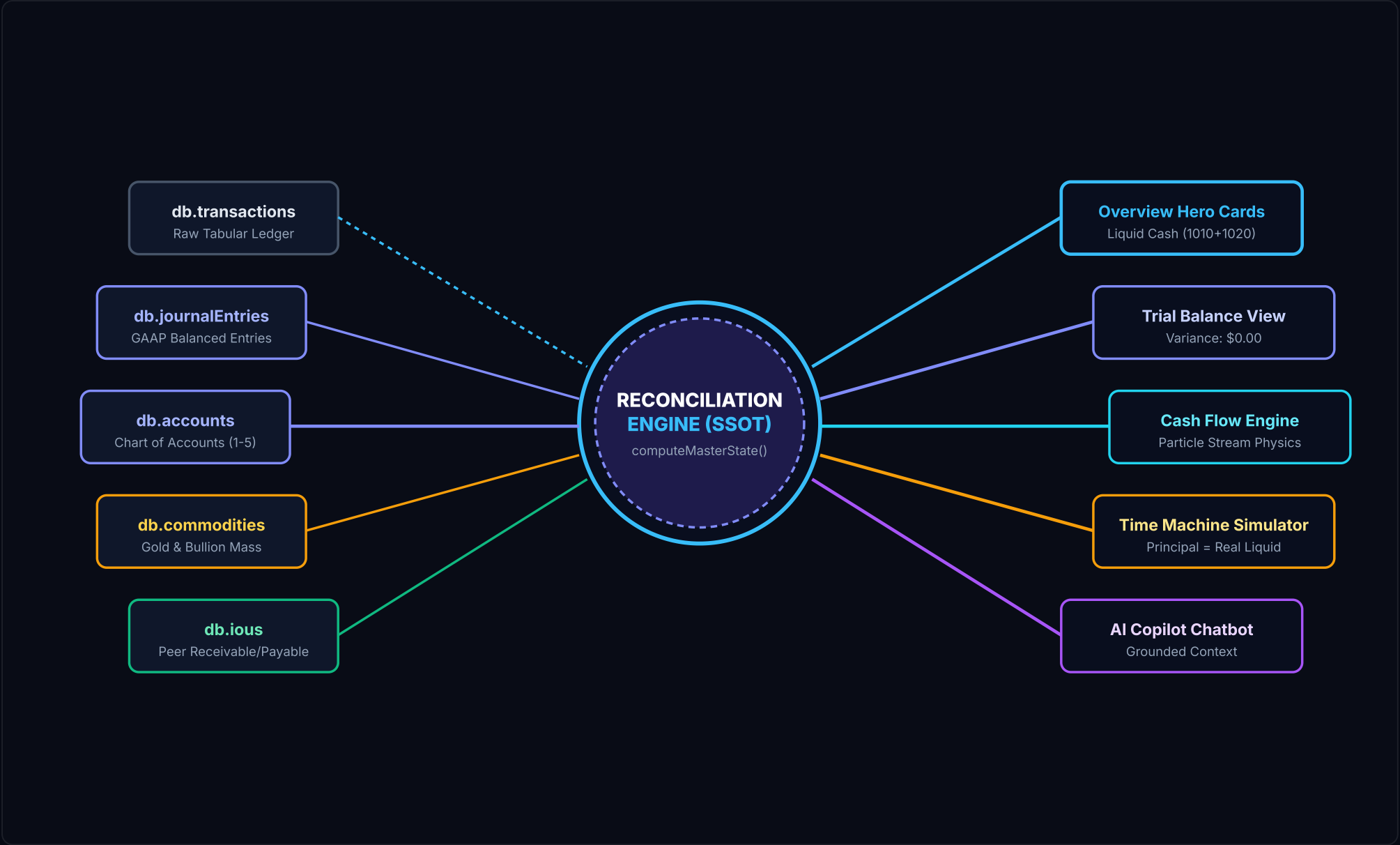


FIG. 03 – RADIAL RECONCILIATION & SYNCHRONIZATION GRAPH
One Single Source of Truth Feeding 15 Autonomous Financial Surfaces

DEXIE.JS REACTIVE OBSERVABLES: `useLiveQuery()`



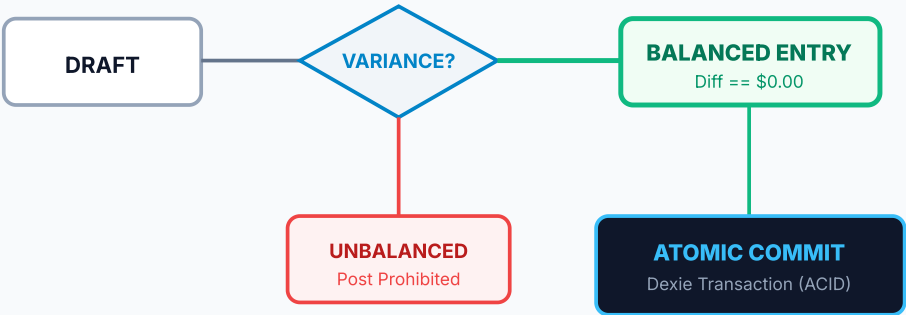
MATHEMATICAL IDENTITY

$$\sum \text{DEBITS} \equiv \sum \text{CREDITS}$$

$$\text{VARIANCE LIMIT: } |\sum \text{Dr} - \sum \text{Cr}| < 0.001$$

Every single financial transaction posted to the system—whether via manual entry, voice command, or multimodal receipt OCR—is converted into a balanced double-entry journal record before entering the ledger.

FIG. 04 - DOUBLE-ENTRY COMMIT STATE MACHINE



RUNTIME EVIDENCE

FIG. 05 — General Ledger & Trial Balance Module

The screenshot shows a web application interface for the General Ledger & Trial Balance Module. The browser address bar shows 'aura-finance-silk.vercel.app/ledger'. The interface includes a header with a home icon, 'Pakistani ...', 'PK PK', 'Engli...', and a search bar. Below the header are three tabs: 'General Journal & OCR Scan' (active), 'Trial Balance Ledger', and 'Balance Sheet & In'. The main content area is titled 'Double-Entry General Journal' with a subtitle 'CPA-Grade Double-Entry Validation Engine. Automatic atomic posting to Chart of Accounts.' It features a 'Posting Date' field set to '26-09-2026' and a 'Narration & Memo *' field with the placeholder 'Enter transaction business rationale...'. Below these is a table with two columns: '#', 'ACCOUNT TITLE & CODE', and 'DEBIT (\$DR)'. The table contains two entries: 1. '1020 — Bank Operating Account (ASSET)' with a debit of 1500, and 2. '4010 — Sales / Consulting Revenue (REVENUE)' with a debit of 0.00. At the bottom of the table, there is a '+ Add Split Line' button and a summary row showing 'TOTAL DEBITS \$1,500.00' and 'TOTAL CREDITS \$1,500.00', followed by a green 'Balanced (Variance \$0.00)' status. Below the table is a section titled 'Posted General Journal Entries (5)' with a list of entries, including 'JE-2026-0005' dated '2026-09-15' with a 'MANUAL' entry type and a description 'Acquisition of 1 Tola 24K Minted Gold Bar bullion asset reserve'. At the bottom of the interface, there are two status bars: '01 DEBITS / CREDITS VALIDATED' and '02 ZERO CENT VARIANCE'.

FIG. 06 – COMPREHENSIVE UML USE CASE DIAGRAM

Four Specialized Personas Interacting with Autonomous Sub-Systems

ACTOR COVERAGE: 100% MODULE CAPABILITIES

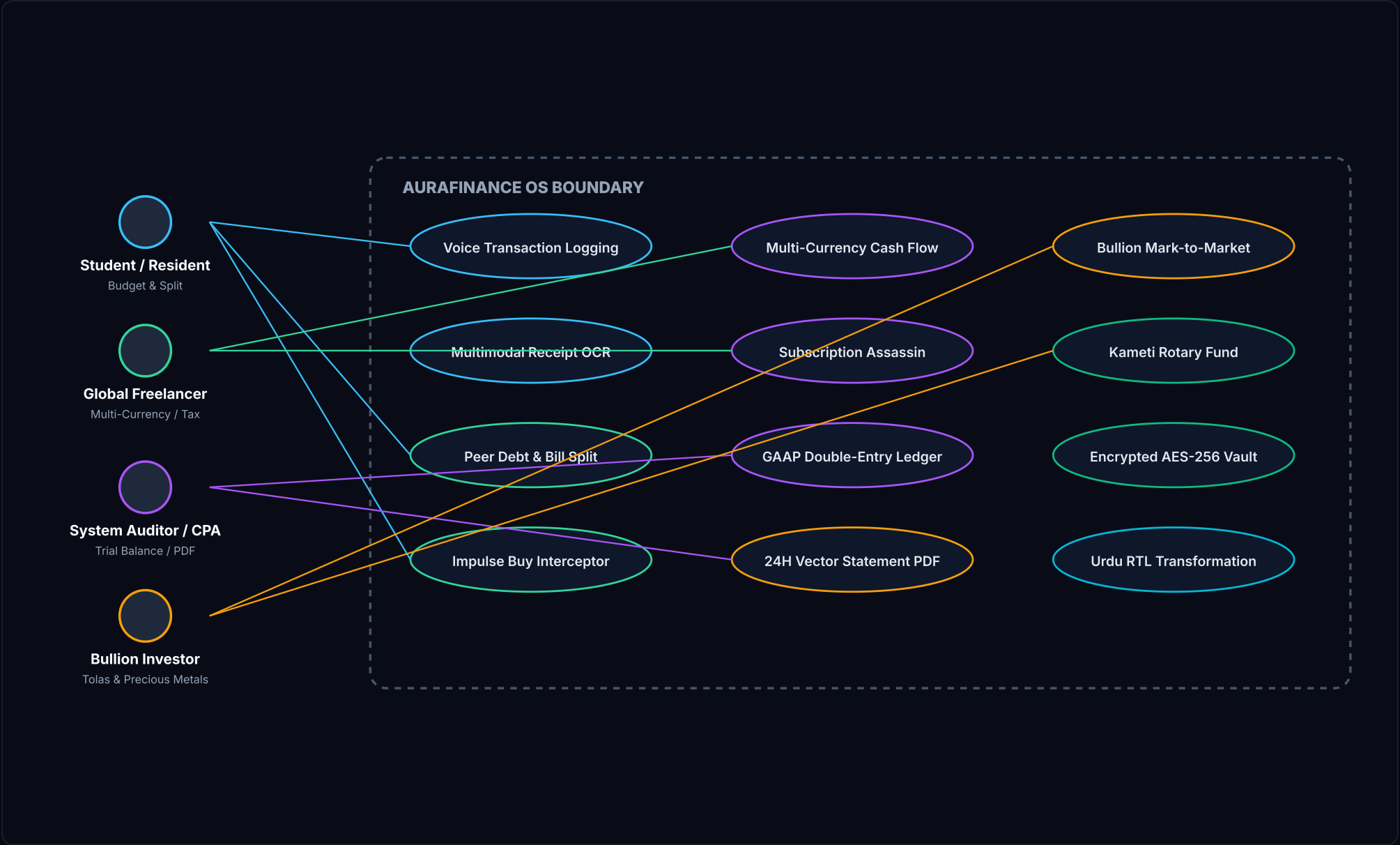


FIG. 07 – INTERACTIVE ONBOARDING & AUDIT TIMELINE
From First Paint to Complete Cryptographic Statement Audit

TIME-TO-VALUE: IMMEDIATE <15 SECONDS

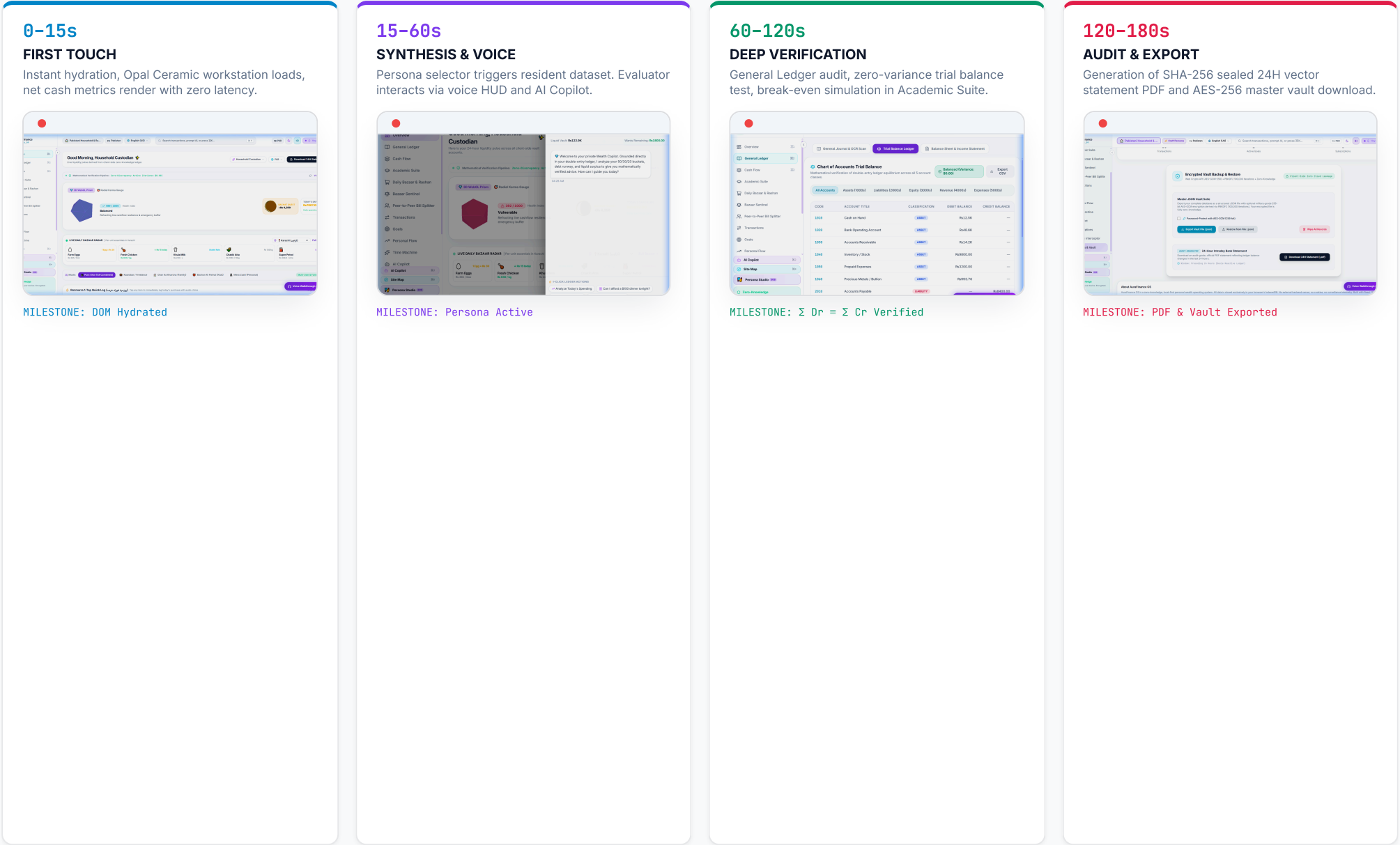


FIG. 08 – OCR TO DOUBLE-ENTRY TRANSACTION PIPELINE

From Physical Receipt Pixels to Balanced Double-Entry Journal Entry

ENGINE: GEMINI 2.0 FLASH VISION + DETERMINISTIC COA MAPPER

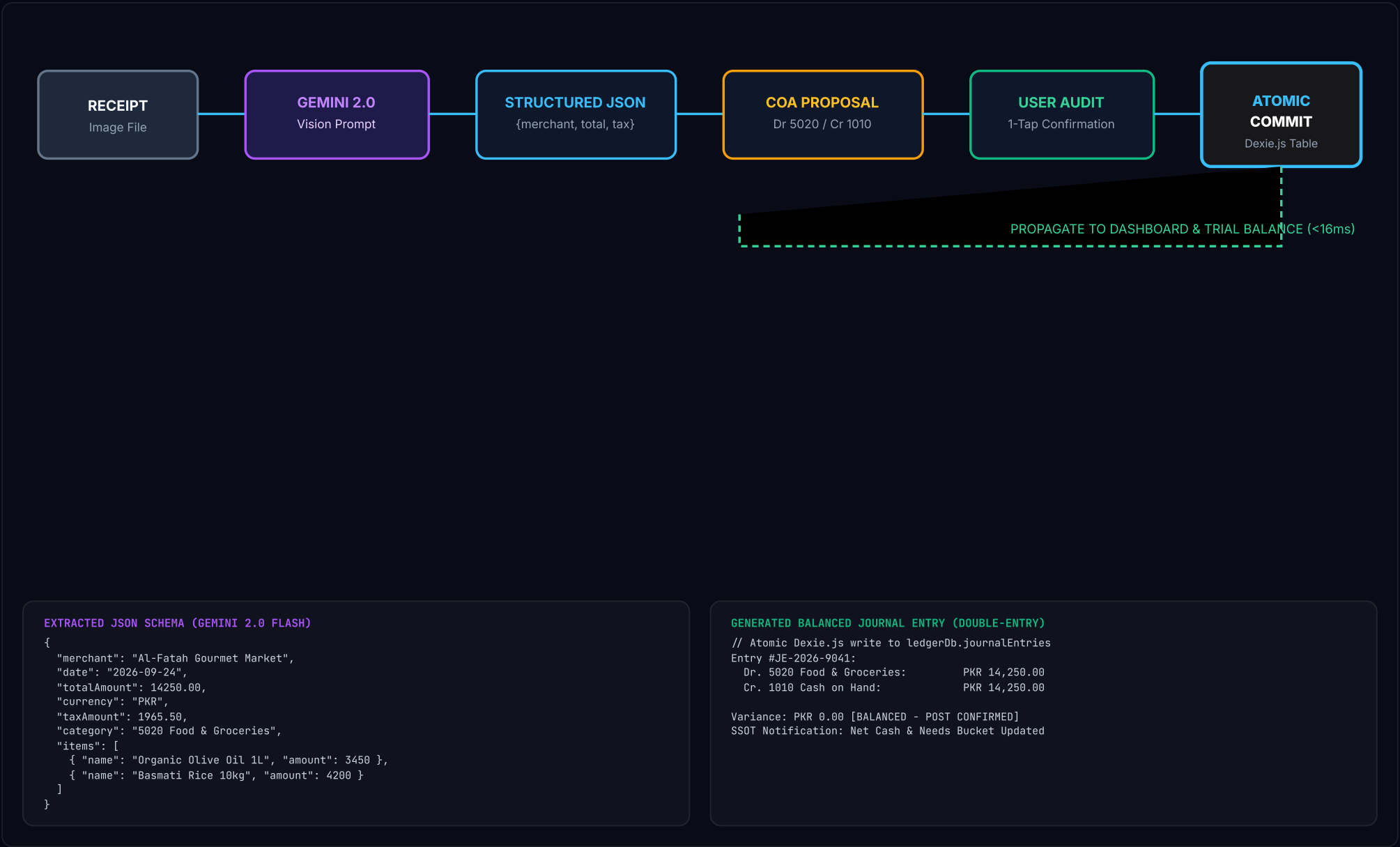
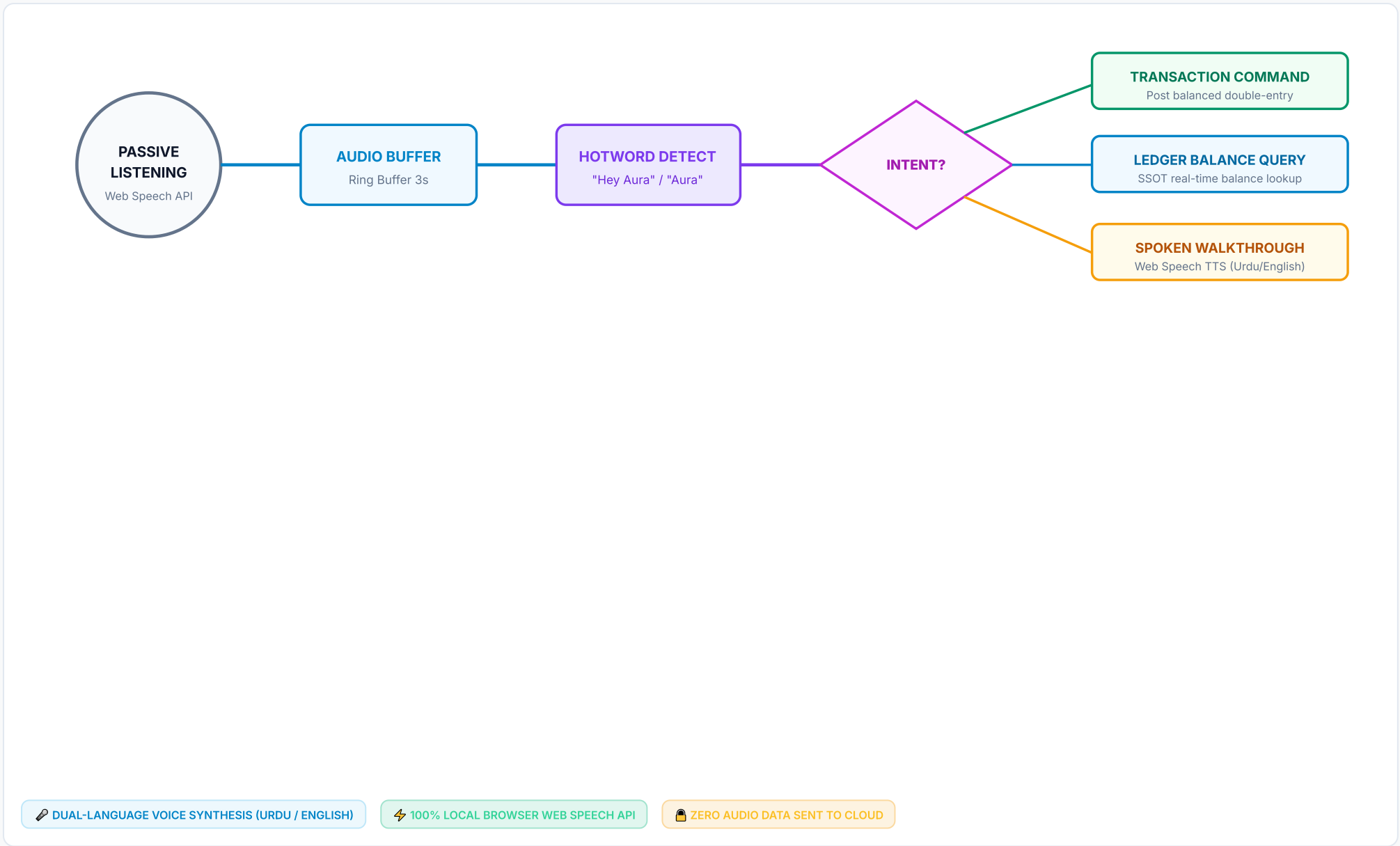


FIG. 09 – VOICE ASSISTANT FINITE STATE MACHINE

From Audio Buffer to Natural Voice Walkthrough (بول کر سنیں)

LATENCY: <180MS HOTWORD DETECT



AUDIT PIPELINE

FIG. 10 — Client-Side Vector PDF Statement Generation

Generates audit-grade PDF statements directly in the browser memory using jsPDF and jsPDF-autotable, sealed with a real-time SHA-256 cryptographic hash of all included records.

- 01
- Evaluator initiates 24H Statement download
- 02
- Query ledgerDb.journalEntries for preceding 24h
- 03
- Calculate Opening Balance, Total Debits & Credits
- 04
- Compute SHA-256 Digest over canonical JSON payload
- 05
- Embed Digital Seal & Stream Vector PDF into Blob

DIGITAL SEAL: SHA-256 e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

RUNTIME EVIDENCE

FIG. 11 — Statement Generation Modal & Verification Deck

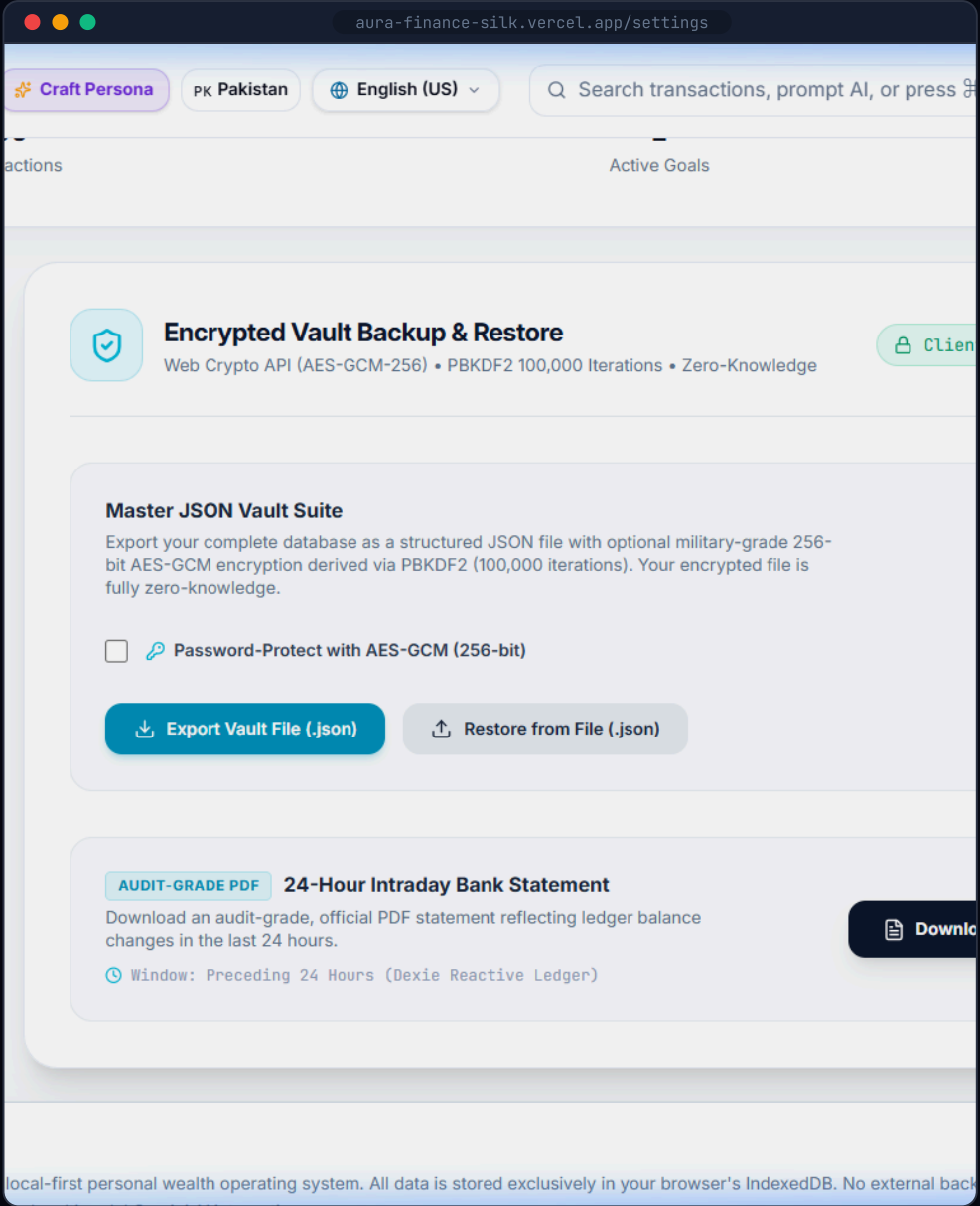


FIG. 12 – VAULT ENCRYPTION & DECRYPTION SEQUENCE

Hardware-Accelerated Web Crypto API Key Derivation & Cipher Pipeline

SPEC: NIST SP 800-132 · 100,000 PBKDF2 ROUNDS

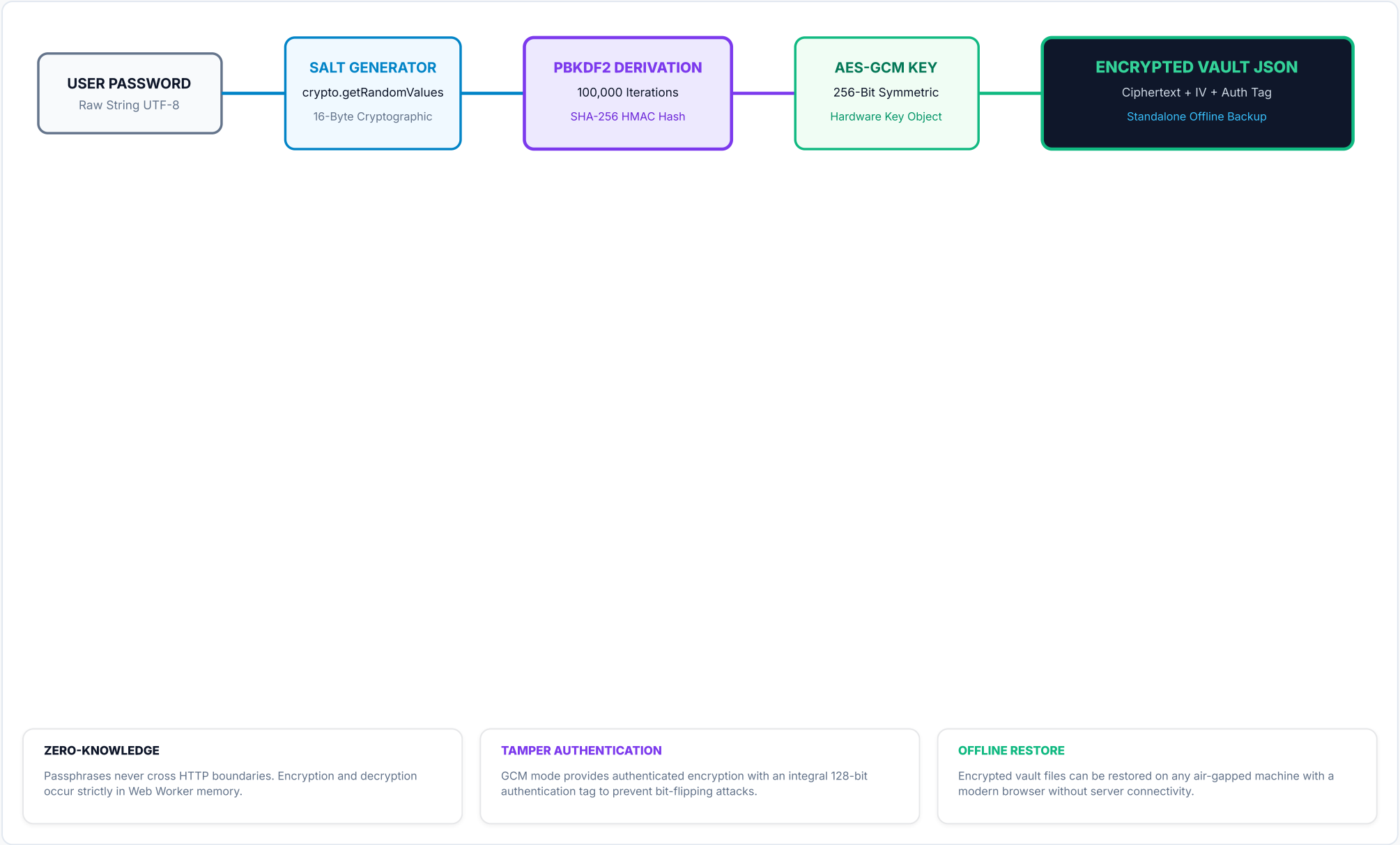


FIG. 13 – DEXIE.JS DATA ENTITY COLLECTIONS

Seven Normalized Local Collections with Reactive Observable Indices

DEXIE VERSION: 4.4.0 · ZERO SERVER SYNCHRONIZATION DEPENDENCY

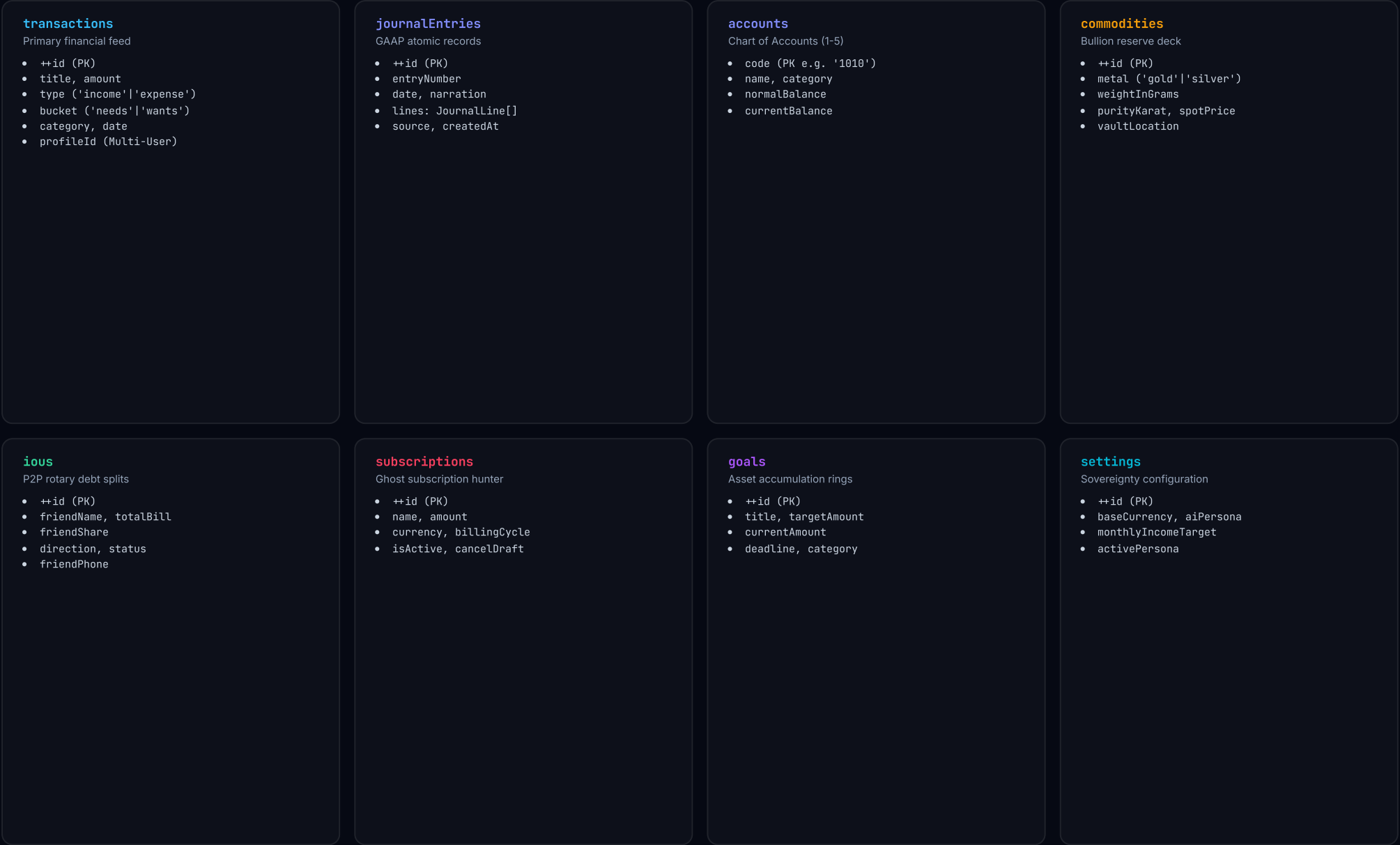
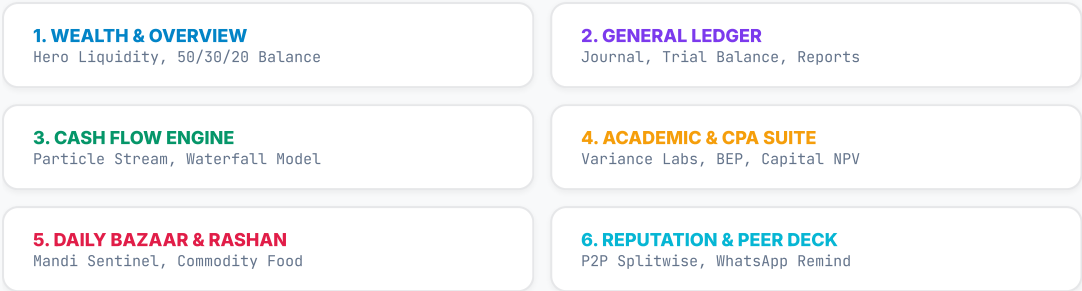


FIG. 14 – SYSTEM TOPOLOGY SPREAD

Six Clustered Financial Workspaces with Keyboard Shortcuts



NAVIGATION ATLAS

Visual System Topology Overlay (CMD+K)

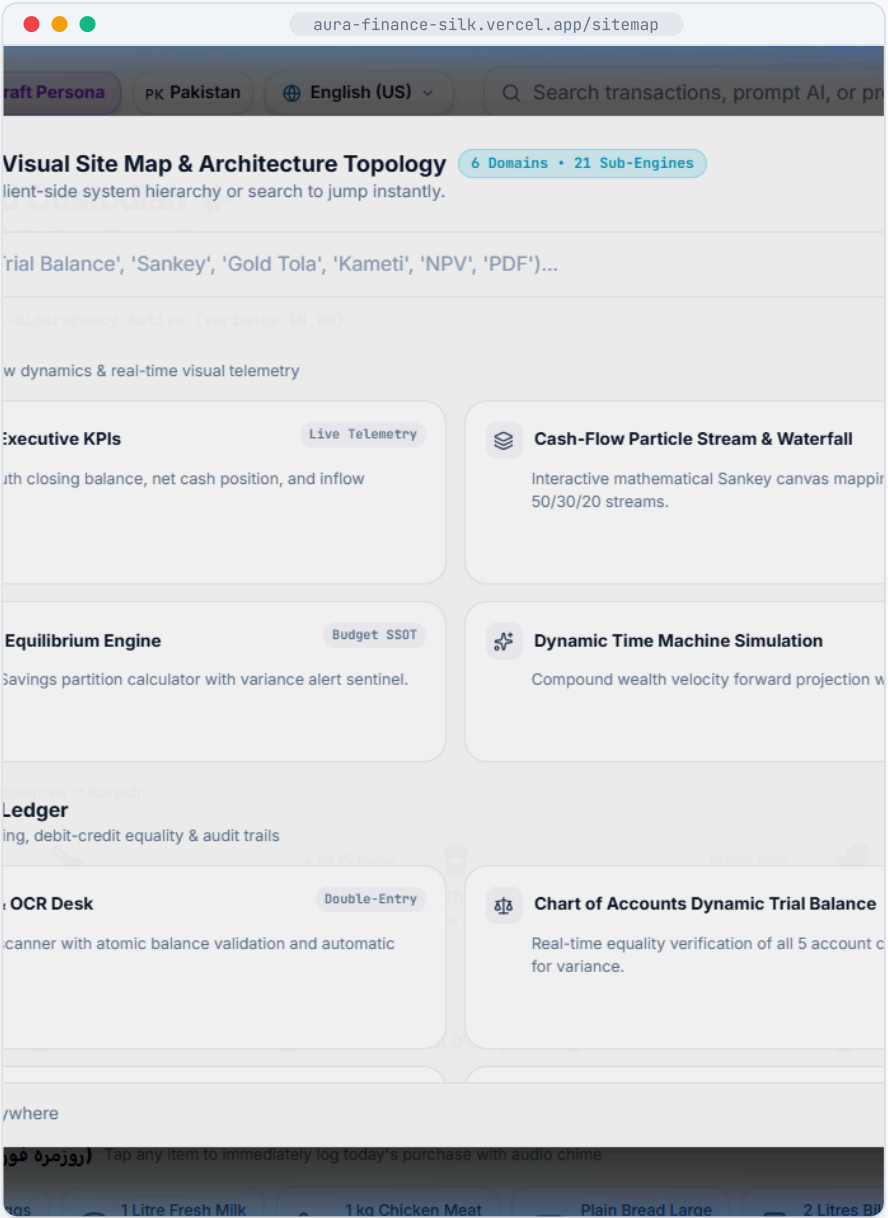


FIG. 15 – BULLION VALUATION & UNIT CONVERSION PIPELINE
Grams, Troy Ounces, and South Asian Tola Calculations

SOUTH ASIAN METRIC STANDARD: 1 TOLA ≡ 11.6638 GRAMS

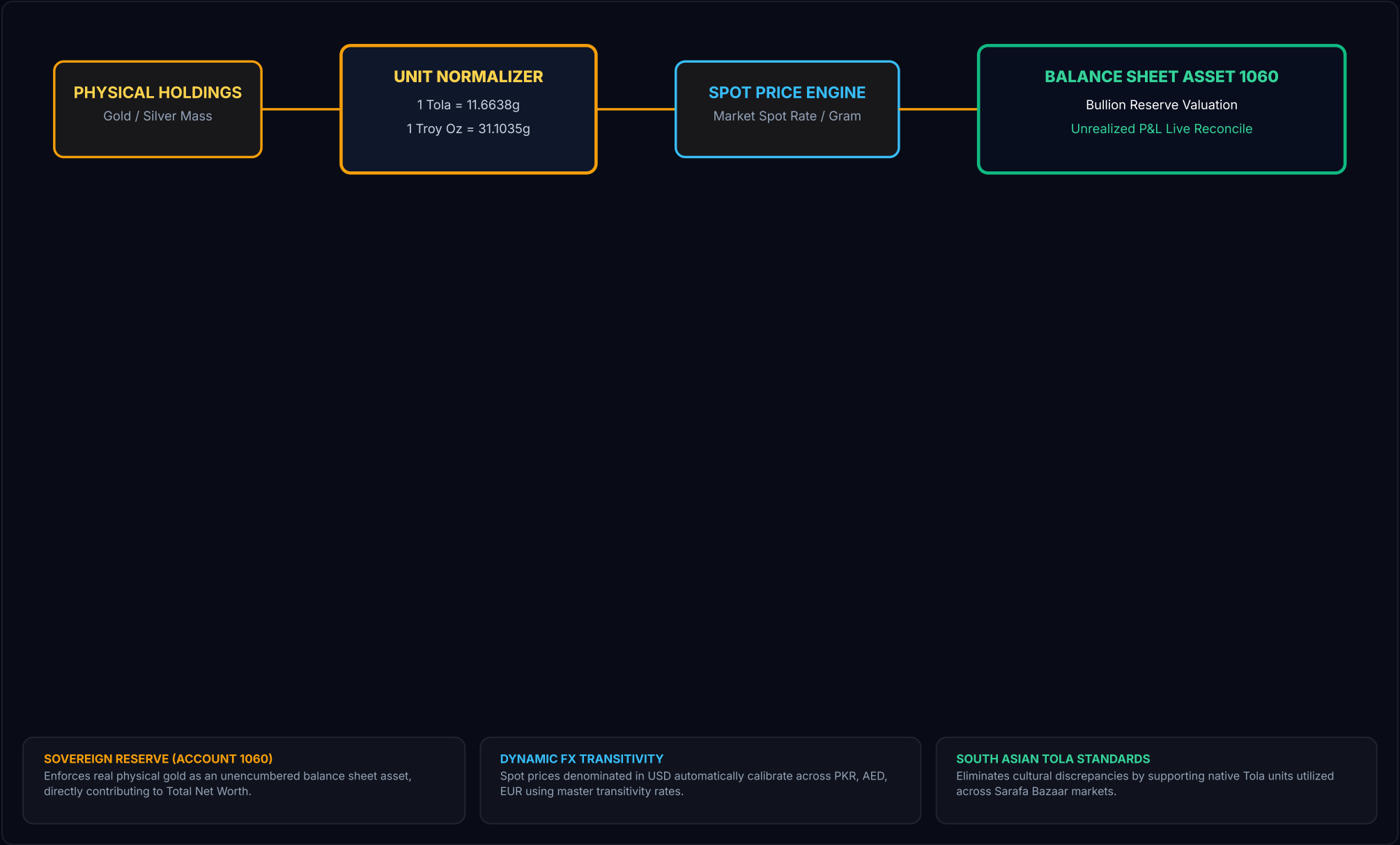
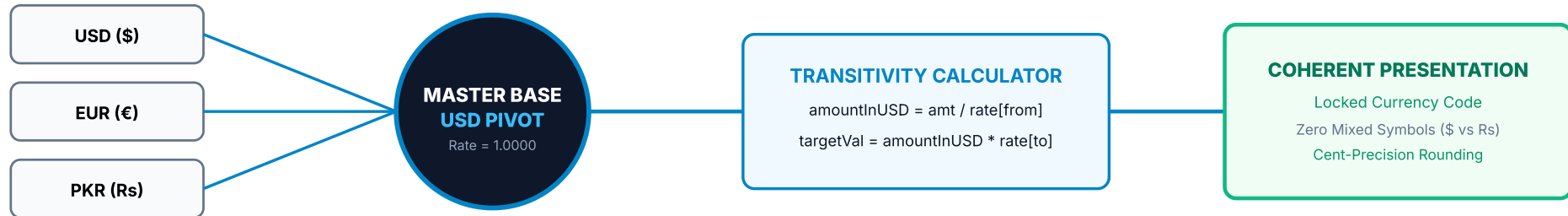


FIG. 16 – MASTER FX RATE TRANSITIVITY ARCHITECTURE

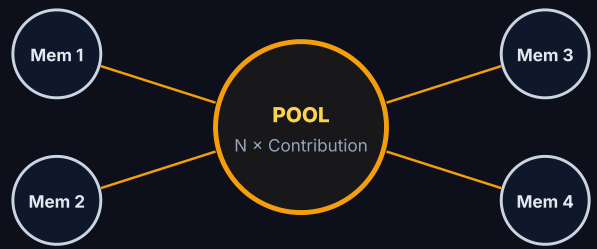
All Ledger Records Unified to Base Currency without Floating-Point DriftMATHEMATICAL LAW 5: `PRECISION ROUND(AMOUNT_USD * RATE_TARGET, 2)`

Multi-Currency Law: It is strictly forbidden for Overview to show \$ while General Ledger shows Rs or Time Machine shows €. When the base currency is toggled, all 15 views re-render with identical mathematical normalization.

MODULE SPECIFICATION

Kameti / Chit Fund Rotary Pool Architecture

A formal mathematical implementation of communal rotating credit associations (ROSCAs), popular across South Asia, the Middle East, and Latin America.

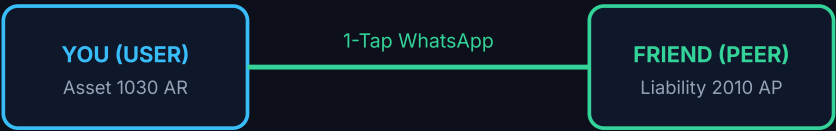


ACCOUNT BINDING: Asset 1050 Rotary Receivables / Liability 2030 Rotary Due

MODULE SPECIFICATION

Peer-to-Peer Debt & 1-Tap WhatsApp Settle

Split shared dining, groceries, and travel bills with zero-backend local storage and instant WhatsApp payment links.



SYNCHRONY LAW: Unsettled IOUs directly scale Net Worth asset totals.

FIG. 17 — DOM DIR="LTR" VS DIR="RTL" MIRRORING ARCHITECTURE

Localization as System Behavior: Mirrored Layouts & Native Financial Vocabulary

SUPPORTED: ENGLISH, URDU (اردو), ARABIC (العربية), ROMAN URDU

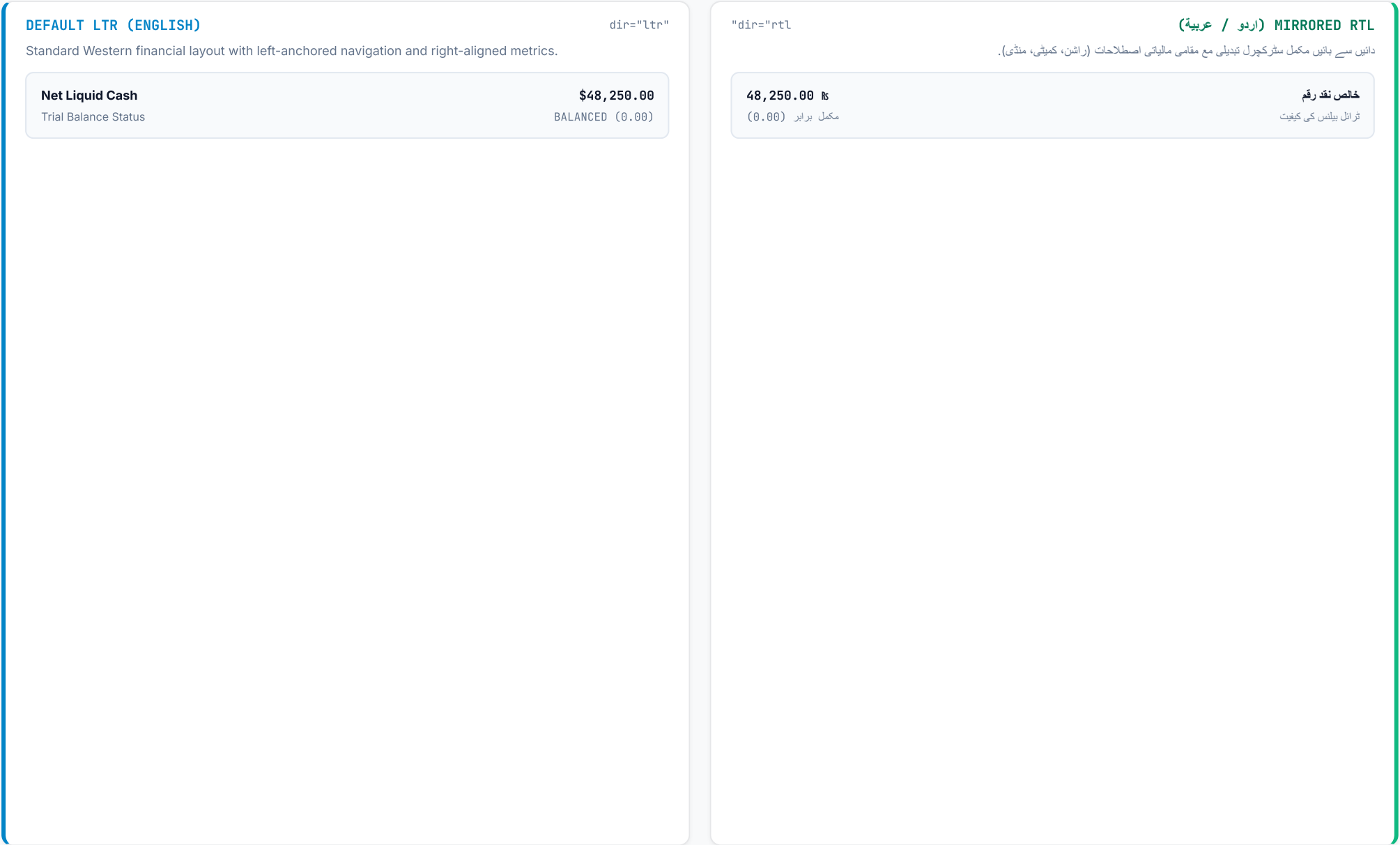


FIG. 18 — TRANSACTION TRANSITION STATE GRAPH

Formal State Machine Enforcing ACID Durability in IndexedDB

DETERMINISTIC VALIDATION: ZERO UNBALANCED WRITES

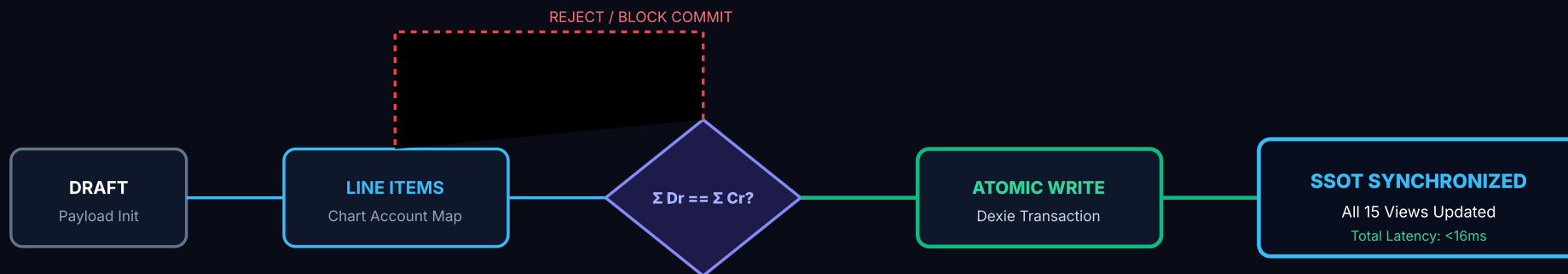


FIG. 19 — SYNCHRONIZED RECONCILIATION PROOF MATRIX

Posting a \$1,200 Revenue Transaction Propagates Instantaneously Across All 15 Modules

REACTION BENCHMARK: SUB-FRAME (<16MS)

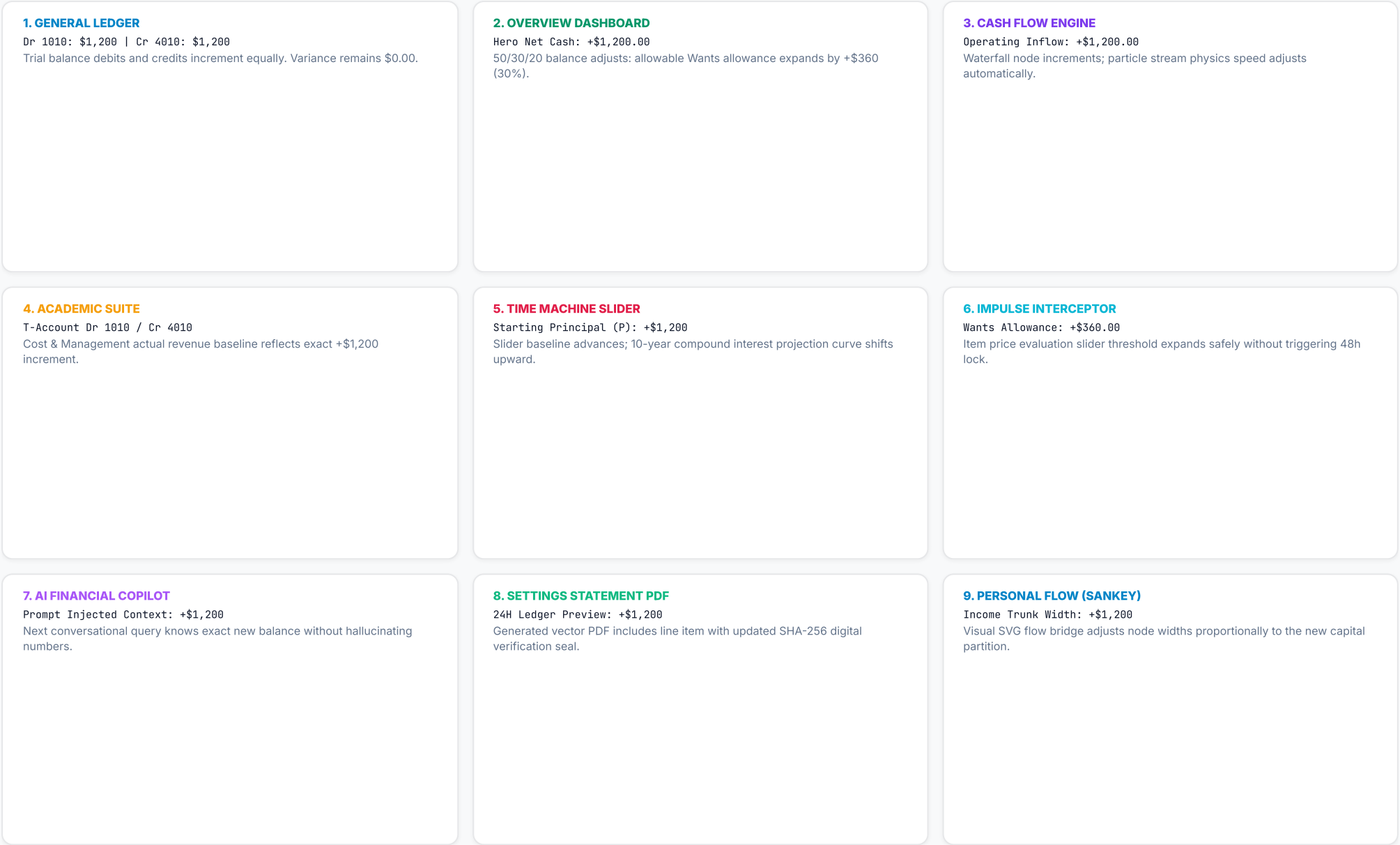


FIG. 20 – STRICT TRUST BOUNDARY SEPARATION

Explicit Visual Contrast: What Stays In the Browser vs. What Crosses the Perimeter

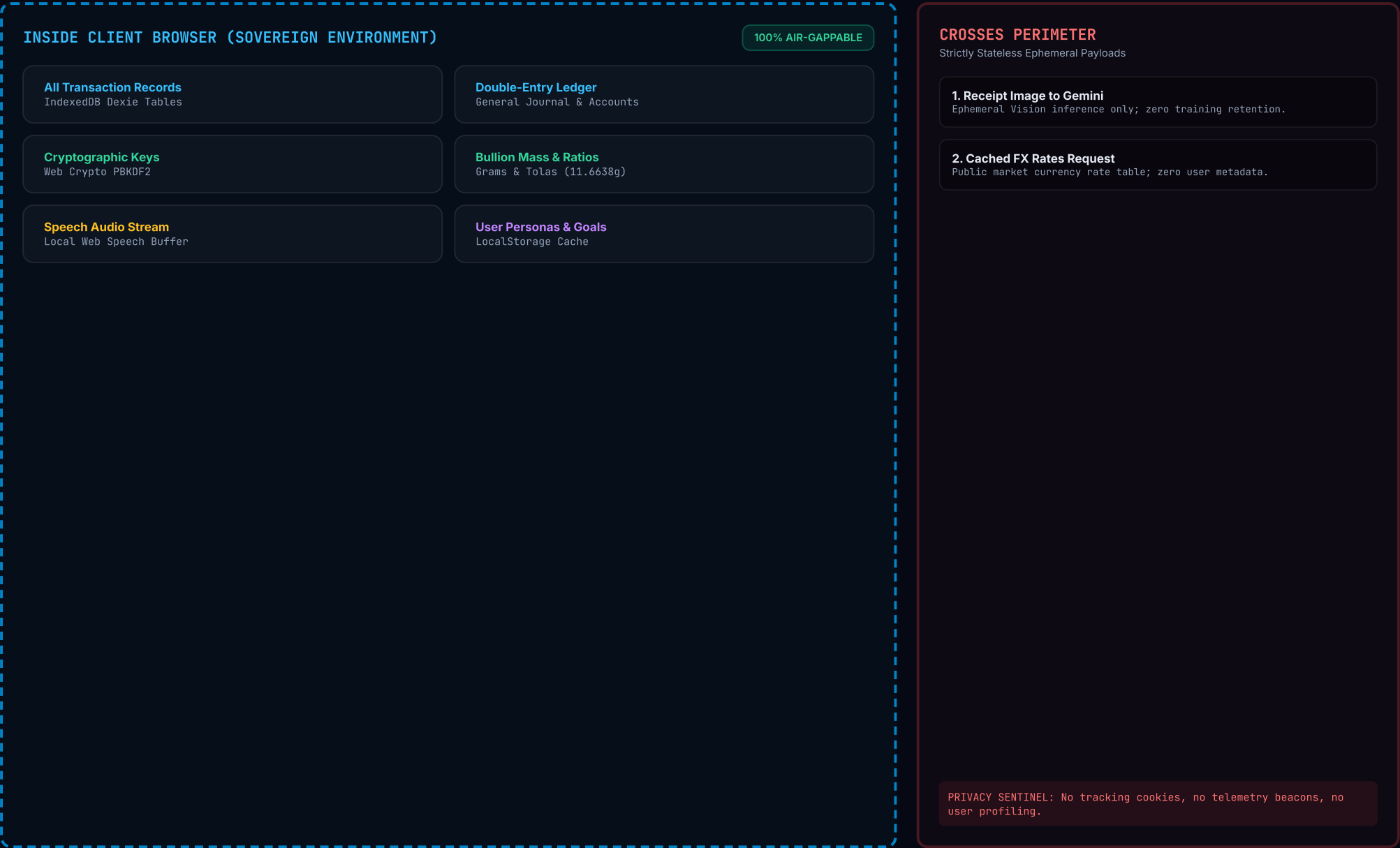


FIG. 21 — PERFORMANCE TELEMETRY BENCHMARKS

Hardware-Agnostic Efficiency Tested Down to Legacy Dual-Core Systems

SPECIFIED IN OFFICIAL ARCHITECTURE DOCUMENTATION

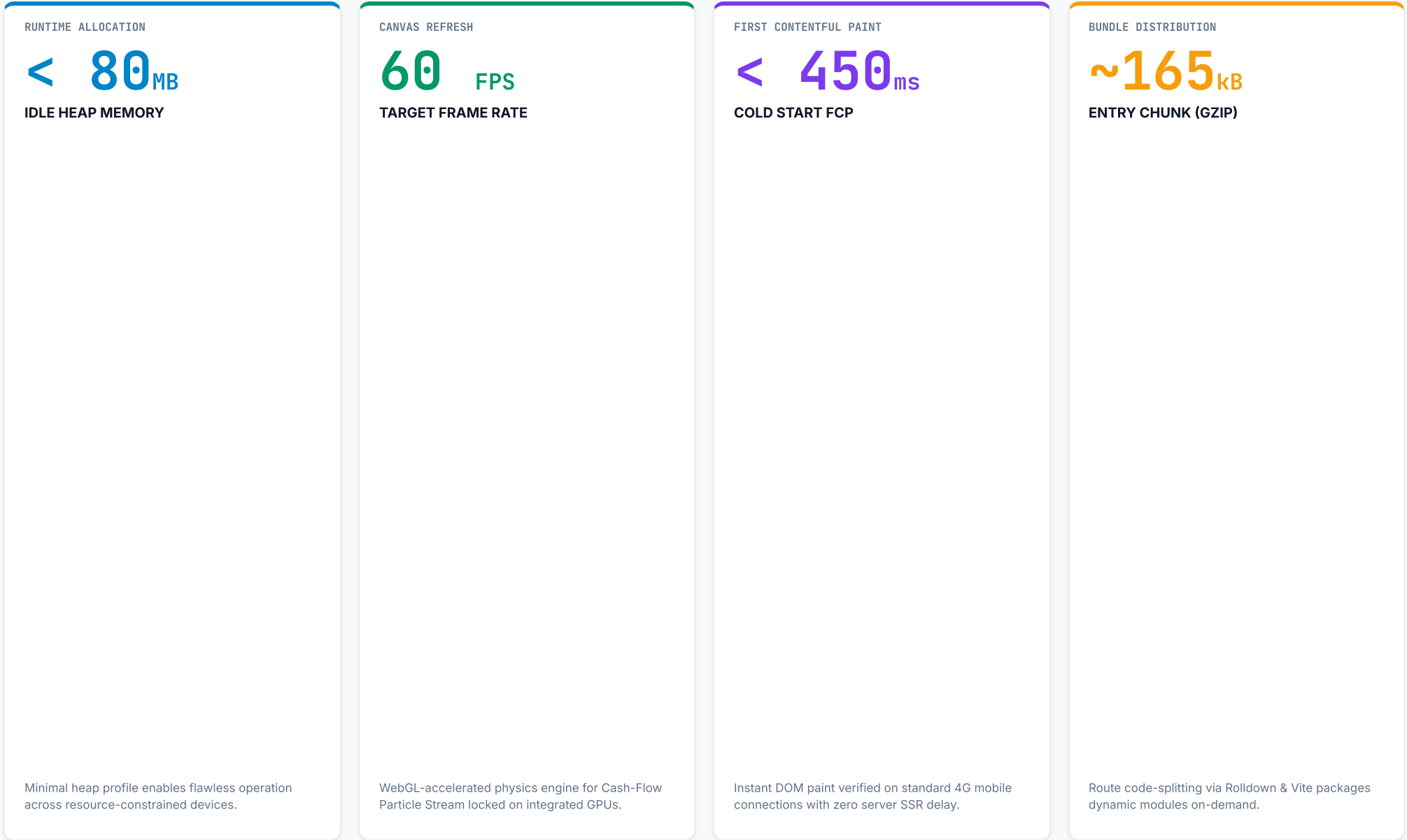


FIG. 22 – EVALUATION PERSONA DOSSIER

Turnkey Real-World Datasets Designed for Comprehensive Jury Audit

1-TAP SYNTHESIS VIA PERSONA ARCHITECT STUDIO

 SYSTEM AUDITOR / CPA High-Frequency Trial Balance Audits standard double-entry journal, verifies zero-variance trials, and generates SHA-256 sealed statements. MODULES: Ledger, Reports, PDF	 GLOBAL FREELANCER Cross-Border USD/EUR/PKR Logs client payments in USD, manages local living expenses in PKR, and tracks tax deductions. MODULES: Multi-Currency, Cash Flow	 MEDICAL RESIDENT / STUDENT Tight Living Stipend Evaluates daily food costs in Mandi Sentinel, tracks shared apartment bills via P2P Splitter. MODULES: Bazaar, IOUs, Interceptor	 BULLION INVESTOR Hard Asset Preservation Monitors physical gold bars in Tolas, tracks spot prices, and models 20-year compound wealth. MODULES: Bullion Vault, Time Machine
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FIG. 23 — UNIFIED SYSTEM ARCHITECTURAL BLUEPRINT

Complete Topology: Ingestion → Intelligence → Persistence → Presentation

COMPLETE CLIENT-SIDE AUTONOMOUS WEALTH SYSTEM



FROM TRANSACTION TO TRUST.

AuraFinance OS proves that personal financial sovereignty does not require cloud surrender. By unifying client-side GAAP double-entry integrity, stateless multimodal intelligence, and hardware-accelerated local encryption, the system delivers an uncompromised wealth operating system built for the next century of computing.

<https://aura-finance-silk.vercel.app/>

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